

MACSE622 – Data Engineering and Visualization LAB

Register Number	25MCB1010	Subject Code	MACSE622
Name of the Student	Patsamatla Bhaavya Ramita	Subject Name	Data Engineering and Visualization
Programme	M.Tech	Slot	L33+L34
Course	CSE (Big Data Analytics)	Semester	2
Date	02-03-26	Exp No	8

Problem Statement:

Consider a “SuperStoreSales” dataset (which has both numerical and categorical values) and draw the following plots in Tableau.

1. Bar plot

- (category,(x axis) sales)
- (region , sales)
- (region (y Axis), product name(y Axis), sales(x axis))
- Region vs discount (x vs y)
- Region vs profit and sales

II. Pie chart

1. Category, profit
2. Category vs Sales

III. Line chat

- X -> time line (order date)
- Y -> sales

IV. Dual plot

- Region vs profit, sales

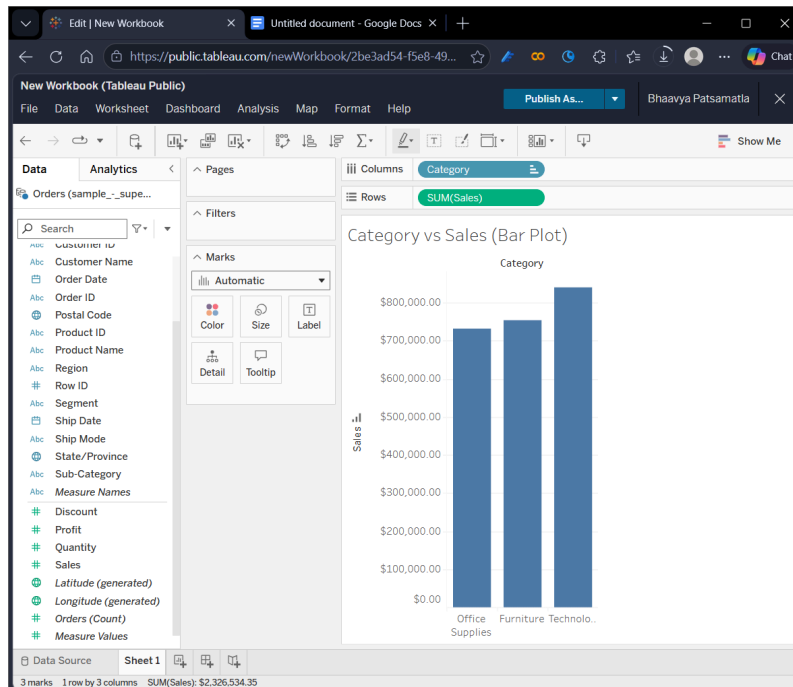
V. Scatter pot

VI. Box plot

VII. Bubble plot

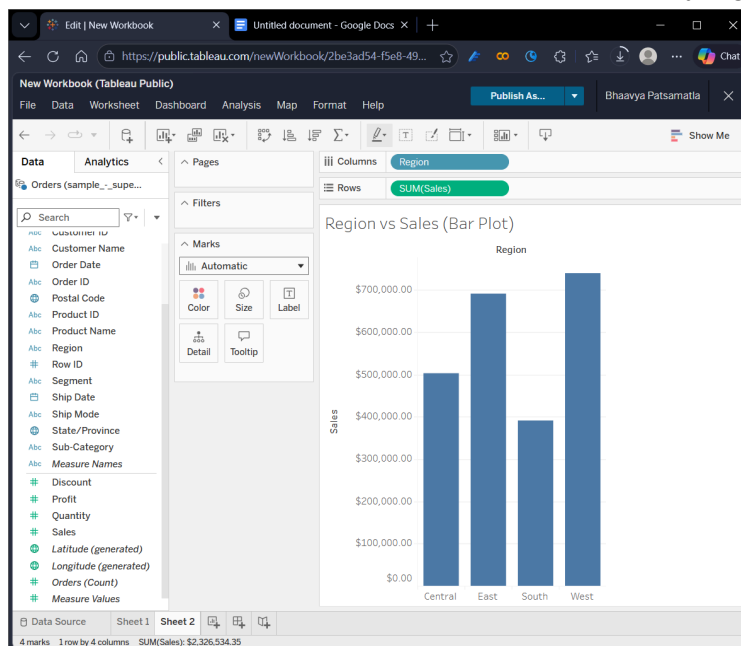
1.1 Category vs Sales (Bar Plot)

This bar chart compares total sales across product categories. Technology shows the highest sales, followed by Furniture and Office Supplies. This indicates that Technology products contribute the most to overall revenue.



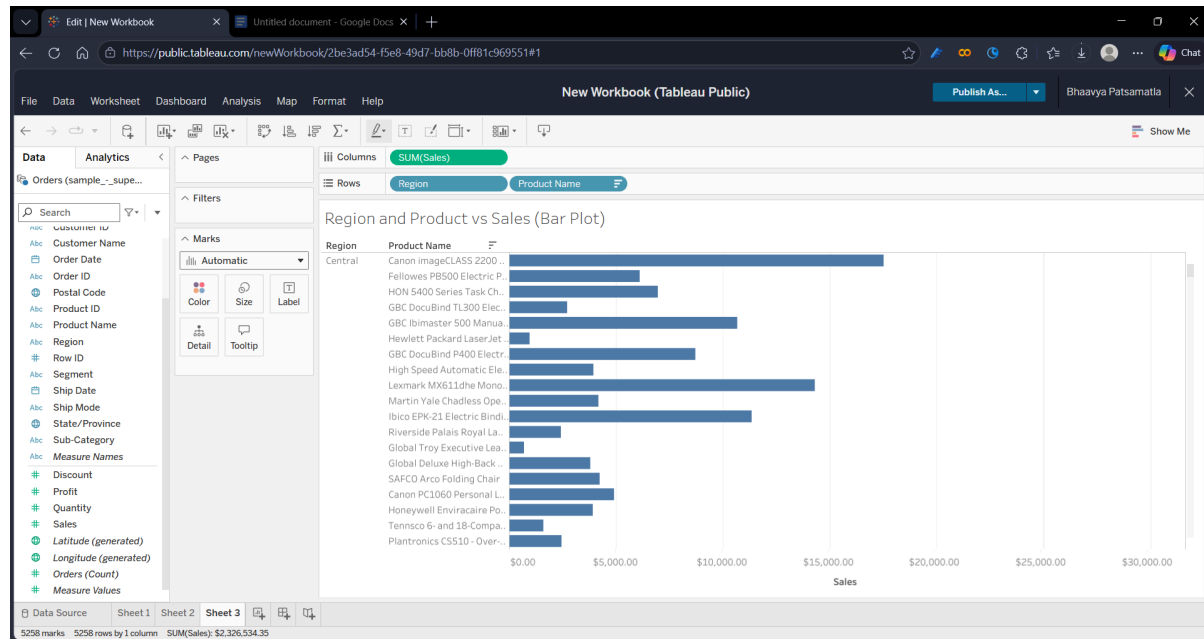
1.2 Region vs sales (Bar Plot)

This visualization represents total sales by region. The West and East regions generate higher sales compared to Central and South. It helps identify high-performing geographical areas.



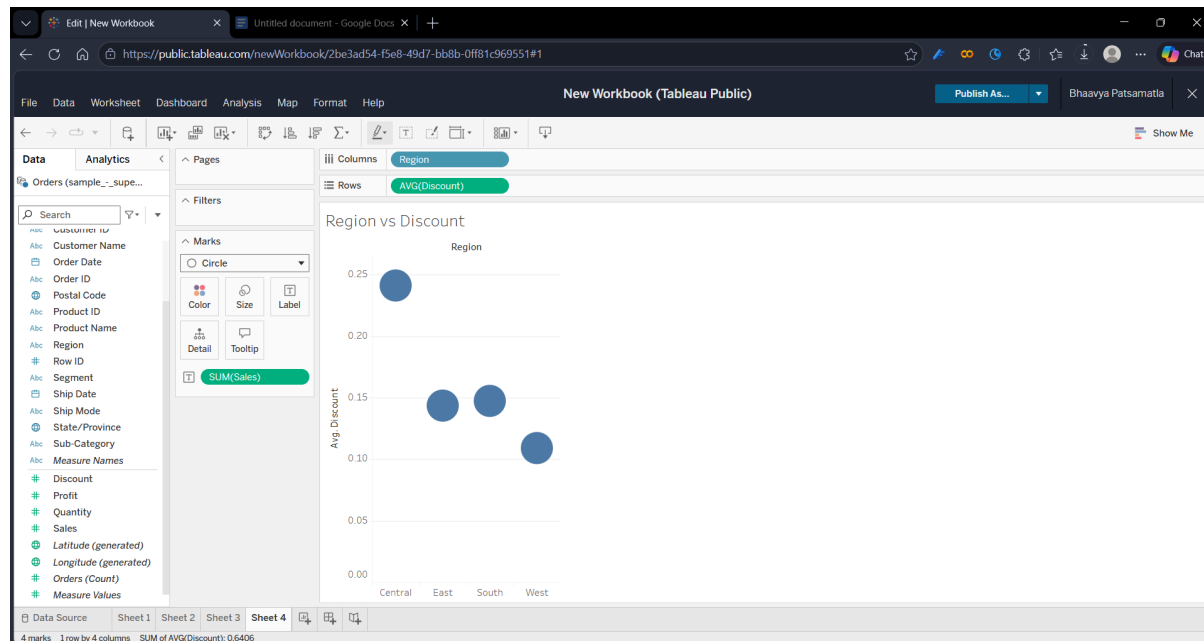
1.3 Region and product vs sales (Bar Plot)

This chart shows sales distribution of products within each region. It helps identify top-performing products in different regions and highlights product-level contribution to regional revenue.



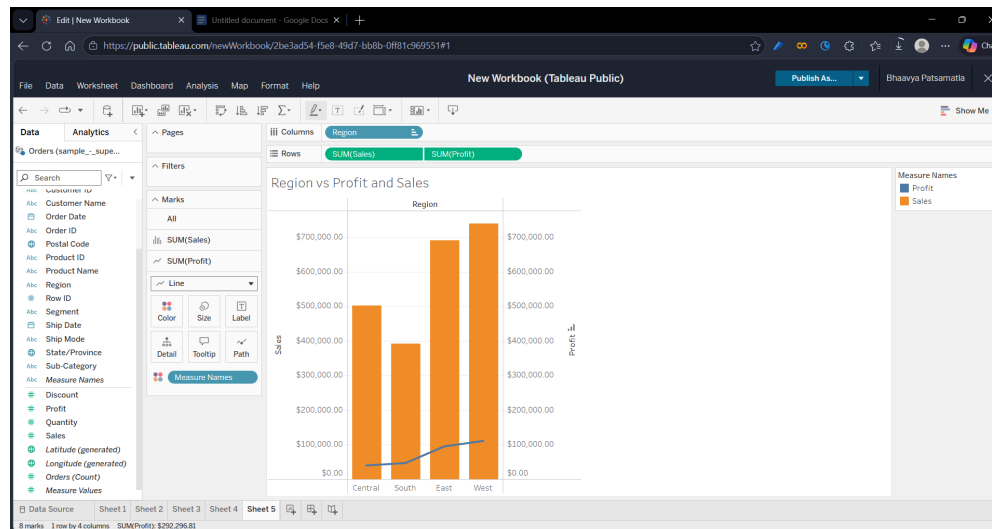
1.4 Region vs discount

This plot displays the average discount offered in each region. It helps analyze pricing strategy differences across regions and evaluate whether higher discounts influence sales performance.



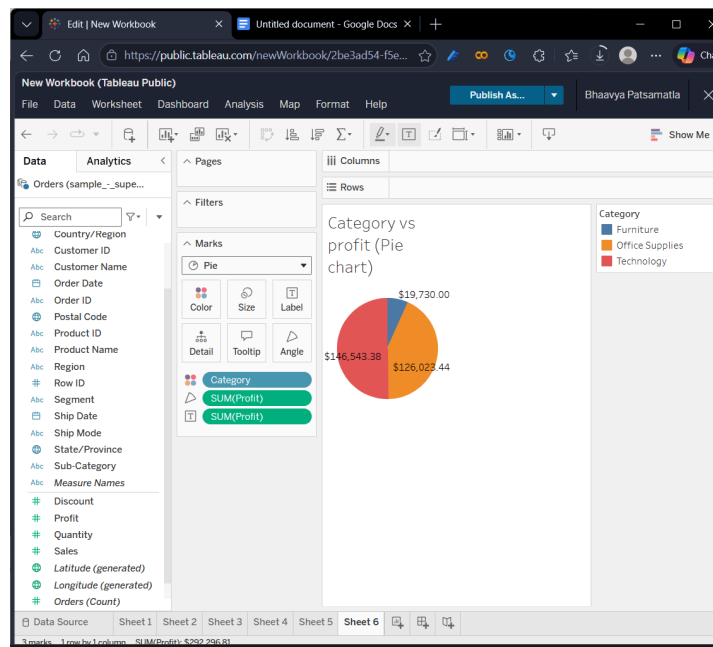
1.5 Region vs profit and sales (Dual Axis)

This dual-axis chart compares sales (bars) and profit (line) across regions. It shows that higher sales do not always guarantee proportionally higher profit, helping evaluate regional profitability.



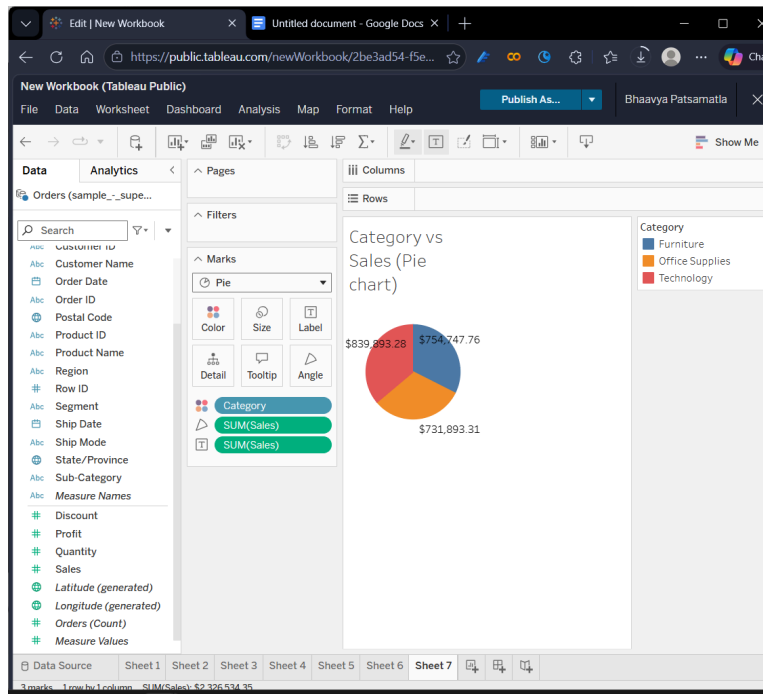
2.1 Category vs profit (Pie chart)

This pie chart illustrates profit contribution by category. Technology contributes the highest profit share, while other categories contribute comparatively less.



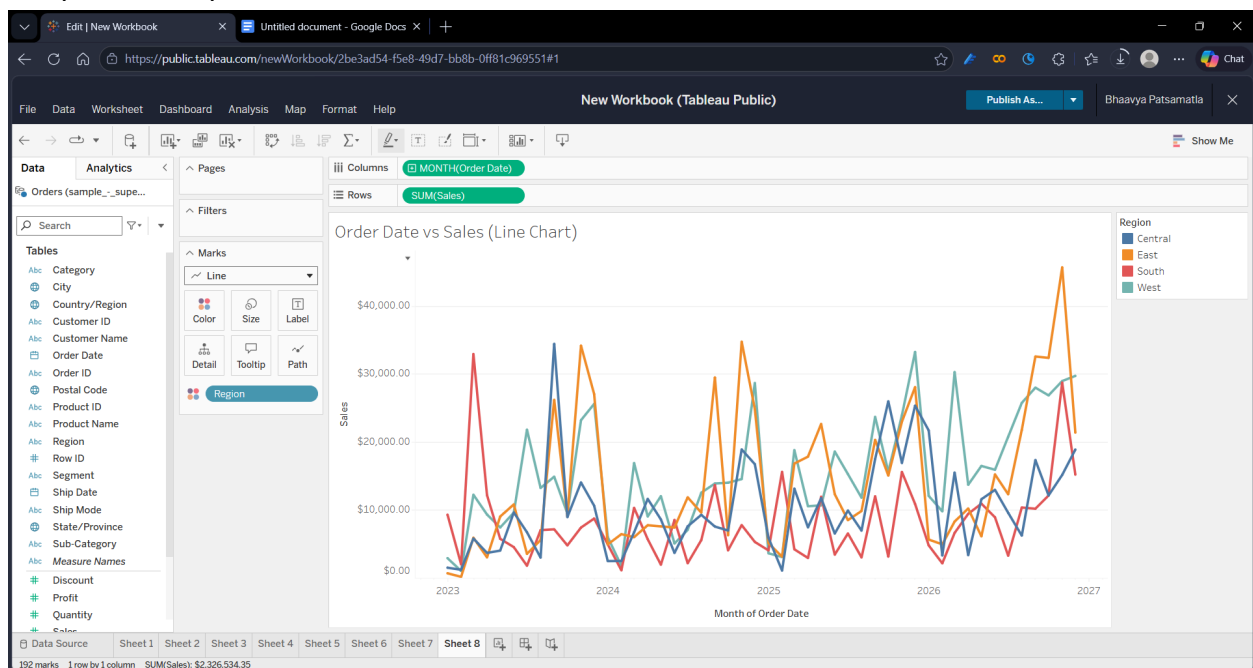
2.2 Category vs sales (Pie chart)

This pie chart represents sales share across categories. It shows the proportional revenue distribution among Furniture, Office Supplies, and Technology.



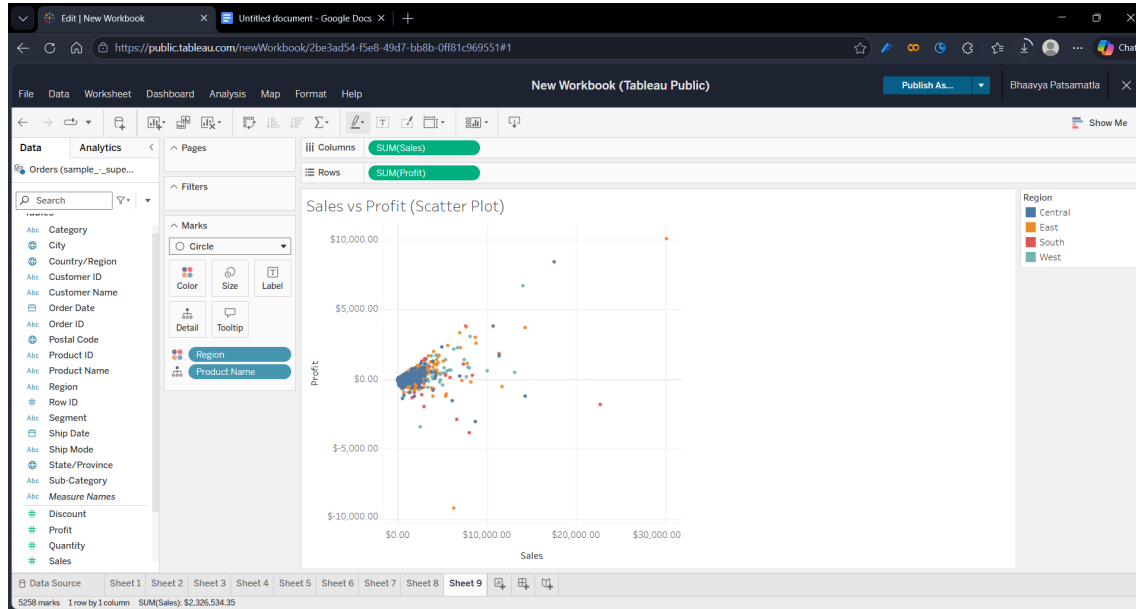
3. Order date vs sales (Line graph)

This line chart shows sales trends over time. It helps identify seasonal patterns, growth trends, and peak sales periods.



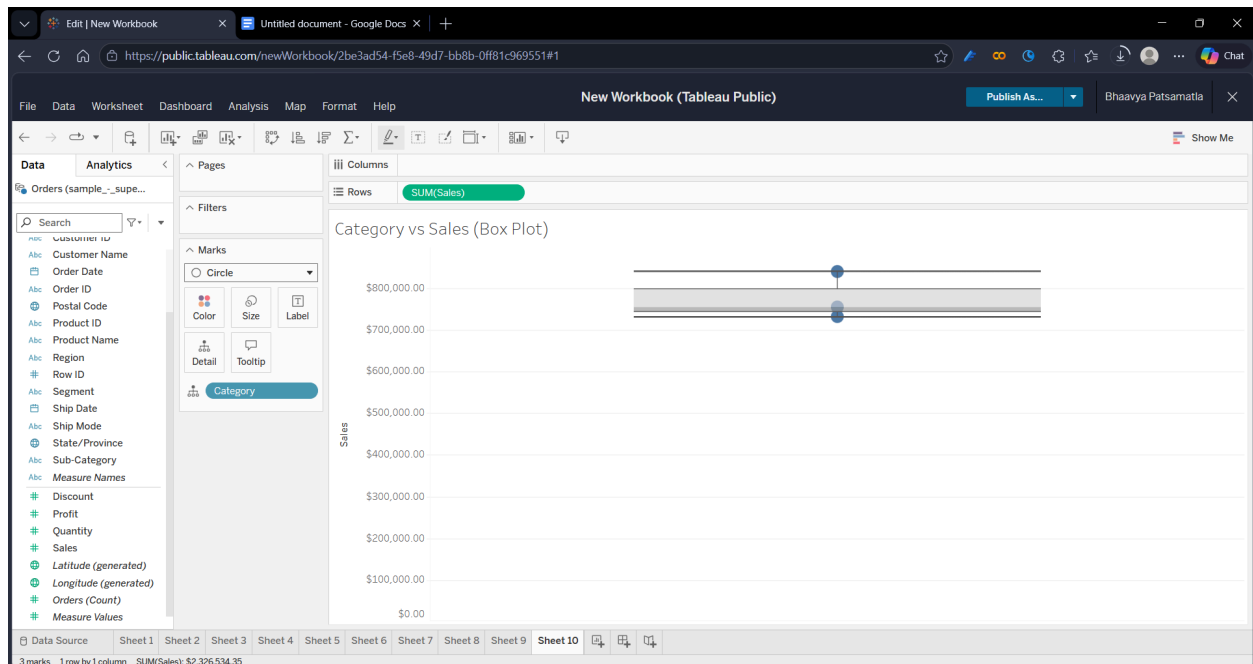
4. Sales vs Profit (Scatter Plot)

This scatter plot analyzes the relationship between sales and profit at the product level. It highlights profitable and loss-making products and helps detect outliers.



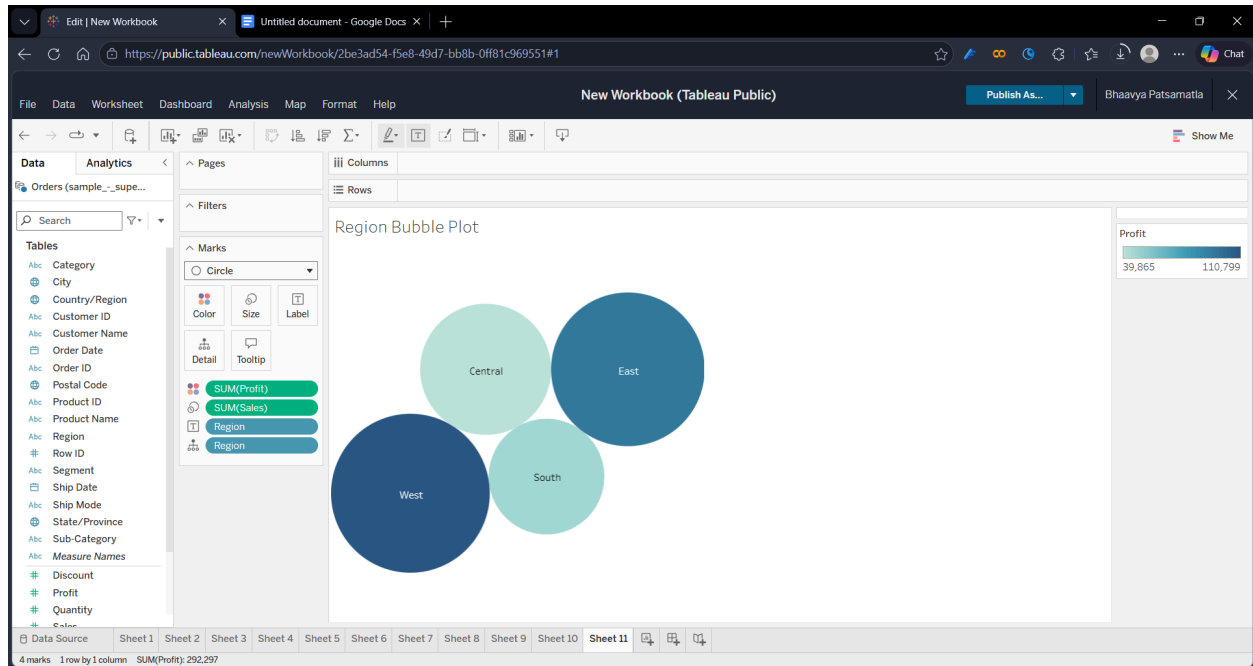
5. Category vs Sales (Box Plot)

This box plot shows sales distribution within each category. It highlights median sales, variability, and outliers, helping understand spread and consistency.



6. Region (Bubble Plot)

This bubble chart represents regional performance, where bubble size indicates sales and color represents profit. It provides a quick visual comparison of overall regional performance.



Submitted By: Patsamatla Bhaavya Ramita 25MCB1010